

**SYSTEMIC FACTORS — HYDROSTATIC/ONCOTIC IMBALANCE**

**TRANSUDATE**

**IF ANY ONE GAUGE FIRES → EXUDATE**

**LOCAL PLEURAL FACTORS ALTERED**

**DISEASE-SPECIFIC TEST SHELF**

**ALL THREE MUST FAIL**

**ALL FAIL → TRANSUDATE**

**PROTEIN RATIO**  
PF/SERUM >0.5

**LDH RATIO**  
PF/SERUM >0.6

**Pf LDH**  
>2/3 ULN

**SPPG >3.1 = CHF TRANSUDATE**  
**CHF stamp**

**SPPG — PLEURAL PROTEIN GRADIENT**  
3.1 g/dL

**SPPG — PLEURAL ALBUMIN GRADIENT**  
SPAG >1.2 = HEPATIC HYDROTIC

**STOP**

**EXUDATE**

**PNEUMONIA ★**

**MALIGNANCY**

**RA**

**NT-proBNP ★**  
>1500 pg/mL  
→ CHF ONLY

**TRIGLYCERIDES**  
>110 mg/dL  
→ CHYLOTHORAX

**ADA**  
>40 IU/L

**GLUCOSE <2**  
<40 mg/dL  
→ RA

**>50% BLOOD Hct**  
→ HEMOTHORAX

**EOSINOPHILS**  
DRUG-INDUCED

**CHIRROSIS**

**NEPHROTIC**

**DIALYSIS**

**CHF ★**

**PSEUDO-EXUDATE — DIURSED CHF**

**DISGUISED SUITCASE**



## 1. Transudate gateway

### WHAT YOU SEE

Blue archway labeled TRANSUDATE with calm patients in line

### WHAT IT ENCODES

Transudate = a protein-poor ultrafiltrate driven by altered systemic hydrostatic/oncotic forces across an INTACT pleura (increased hydrostatic pressure as in heart failure, or decreased oncotic pressure as in cirrhosis/nephrotic syndrome). The pleural surfaces themselves are normal — the fluid is a consequence of a systemic problem.

### BOARD TRAP

A transudate does NOT need an extensive pleural workup (no cytology/biopsy hunt) — treat the underlying systemic disease (e.g., diurese the heart failure). The trap is sending a transudate for malignancy workup or draining it repeatedly instead of optimizing the systemic cause.

## 2. All 3 must fail

### WHAT YOU SEE

Sign 'ALL THREE MUST FAIL → TRANSUDATE'

### WHAT IT ENCODES

By Light's criteria, fluid is a transudate ONLY if ALL THREE criteria are negative (none met). The criteria are deliberately tuned for high sensitivity to catch exudates, so a transudate is a diagnosis of exclusion within this framework.

### BOARD TRAP

Meeting zero criteria = transudate; you cannot call something a transudate if even ONE criterion is positive. The trap is averaging the results or ignoring a single mildly positive criterion — Light's is an OR rule, so one positive flips it to exudate.

## 3. One gauge fires = exudate

### WHAT YOU SEE

Banner 'IF ANY ONE GAUGE FIRES → EXUDATE'

### WHAT IT ENCODES

Light's criteria — fluid is an EXUDATE if ANY ONE of: (1) pleural/serum protein ratio > 0.5, (2) pleural/serum LDH ratio > 0.6, or (3) pleural LDH > two-thirds the upper limit of normal serum LDH. Sensitivity for exudate is ~98%, so it rarely misses an exudate but over-calls (low specificity).

### BOARD TRAP

You do NOT need all three positive — a single positive criterion makes it an exudate. The trap is requiring multiple criteria. Because the rule over-calls exudates (especially in diuresed HF), a 'borderline exudate' clinically consistent with a transudate should prompt an albumin gradient check rather than a malignancy workup.

## 4. Protein ratio >0.5

### WHAT YOU SEE

Gauge: PROTEIN RATIO pleural/serum >0.5

### WHAT IT ENCODES

Light's criterion 1: pleural-fluid total protein / serum total protein > 0.5 indicates exudate. As a standalone clue, a pleural protein > 3.0 g/dL suggests exudate; very high protein (> 4 g/dL) raises suspicion for TB.

### BOARD TRAP

Don't use pleural protein in isolation — it must be a RATIO to serum protein. A patient with low serum protein (e.g., nephrotic) can have a deceptively high ratio. Interpret with the full criteria set and the clinical context.

## 5. LDH ratio >0.6

### WHAT YOU SEE

Gauge: LDH RATIO pleural/serum >0.6

### WHAT IT ENCODES

Light's criterion 2: pleural-fluid LDH / serum LDH > 0.6 indicates exudate. LDH reflects cellular turnover/inflammation; a markedly high pleural LDH (e.g., > 1000) suggests empyema, malignancy, TB, or rheumatoid pleuritis.

### BOARD TRAP

A high pleural LDH with a normal protein ratio still makes it an exudate (any single criterion counts). The trap is dismissing an isolated LDH elevation; instead, a very high LDH narrows the exudate differential toward infection/malignancy.

## 6. Pleural LDH >2/3 ULN

### WHAT YOU SEE

Gauge 'PF LDH >2/3 ULN'

### WHAT IT ENCODES

Light's criterion 3: pleural LDH greater than two-thirds of the upper limit of normal SERUM LDH indicates exudate. This criterion uses the lab's serum LDH reference range (not the patient's serum LDH), so it can flag an exudate even when serum LDH is unavailable.

### BOARD TRAP

Confusing 'two-thirds of the upper limit of normal' with 'two-thirds of the patient's serum LDH' is the trap. Criterion 3 references the normal RANGE; criterion 2 is the patient-specific ratio. Mixing them up changes the classification.

## 7. CHF transudate

### WHAT YOU SEE

Fat suitcase man wearing CHF jersey, 'SPEP >3.1 CHF TRANSUDATE'

### WHAT IT ENCODES

Heart failure is the #1 cause of transudative effusions (often bilateral, right > left). When diuretics have been given and Light's misleadingly suggests exudate, the serum-to-pleural ALBUMIN gradient > 1.2 g/dL (or serum-pleural protein gradient > 3.1 g/dL) reclassifies the fluid as a transudate. Pleural fluid NT-proBNP > 1500 pg/mL also supports a cardiac origin.

### BOARD TRAP

Diuresed HF can falsely meet exudate criteria (pseudoexudate) because diuresis concentrates protein/LDH in the remaining fluid — the trap is launching a malignancy/TB workup. Instead use the serum-pleural albumin gradient > 1.2 g/dL (or NT-proBNP) to correctly call it a transudate.



## 8. Pseudoexudate / diuresed CHF

### WHAT YOU SEE

Disguised suitcase labeled 'PSEUDO-EXUDATE = DIURESED CHF'

### WHAT IT ENCODES

Pseudoexudate = a true transudate (usually HF) that crosses into the exudate range on Light's criteria after diuretic therapy concentrates the protein and LDH. It is the classic Light's-criteria pitfall and is resolved with gradient testing rather than reclassifying the patient's disease.

### BOARD TRAP

Don't chase malignancy/infection in a diuresed HF patient whose fluid 'turned exudative' — check the serum-pleural albumin gradient ( $> 1.2$  g/dL), the protein gradient ( $> 3.1$  g/dL), or pleural NT-proBNP before any extensive exudate workup. Treating the heart failure resolves the effusion.

## 9. Transudate causes lineup

### WHAT YOU SEE

Line of men with bags: CHF, cirrhosis, nephrotic, dialysis

### WHAT IT ENCODES

Transudate causes: heart failure (#1), cirrhosis with hepatic hydrothorax, nephrotic syndrome, hypoalbuminemia, peritoneal dialysis, and (less classically) atelectasis, constrictive pericarditis, and SVC obstruction. The common thread is altered systemic Starling forces with intact pleura.

### BOARD TRAP

Hepatic hydrothorax is usually RIGHT-sided (diaphragmatic defects let ascites pass), and nephrotic effusions are typically BILATERAL transudates. The trap is assuming any cirrhotic/nephrotic effusion needs a pleural workup — treat the systemic cause; tap mainly to exclude spontaneous bacterial empyema in hepatic hydrothorax with fever.

## 10. Exudate gateway

### WHAT YOU SEE

Red archway labeled EXUDATE, distressed patients

### WHAT IT ENCODES

Exudate = fluid produced by LOCAL pleural pathology — increased capillary permeability (inflammation/infection/malignancy) or impaired lymphatic drainage. The pleural surfaces are abnormal, so an exudate always warrants identifying the specific cause.

### BOARD TRAP

An exudate is never 'just an exudate' — it requires a cause-directed workup (cytology, cultures, ADA, etc.). The trap is stopping at the Light's classification; the next step is determining WHY it is an exudate.

## 11. Pneumonia / parapneumonic

### WHAT YOU SEE

Coughing patient with red lungs labeled PNEUMONIA

### WHAT IT ENCODES

Parapneumonic effusion is the #1 cause of exudates; malignancy is #2, followed by PE and TB. The key downstream question for a parapneumonic effusion is whether it is simple (antibiotics) or complicated/empyema (needs drainage).

### BOARD TRAP

A parapneumonic effusion needs pH/glucose/LDH analysis to decide if it is complicated and requires a chest tube — the trap is treating pneumonia + effusion with antibiotics alone without sampling the fluid, missing a developing empyema.

## 12. Malignancy

### WHAT YOU SEE

Patient marked MALIGNANCY

### WHAT IT ENCODES

Malignant effusions are exudates that are often bloody, lymphocyte-predominant, and may have LOW glucose and LOW pH (large tumor burden). Most common primaries: lung, breast, and lymphoma. Diagnosis via cytology (yield ~60% on first tap) and pleural biopsy/thoracoscopy when cytology is negative.

### BOARD TRAP

A single negative cytology does NOT exclude malignancy — the trap is reassuring the patient after one negative tap. Repeat thoracentesis (sending a large volume) raises yield, and persistent suspicion warrants medical/surgical thoracoscopy with pleural biopsy.

## 13. Disease-specific test shelf

### WHAT YOU SEE

Shelf of bottles: NT-proBNP, ADA, glucose, Hct, eosinophils

### WHAT IT ENCODES

Targeted pleural tests: NT-proBNP  $> 1500$  pg/mL → heart failure; ADA  $> 40$  U/L → TB pleuritis; glucose  $< 60$  mg/dL → complicated parapneumonic/empyema, malignancy, TB, or rheumatoid pleuritis; pleural/blood Hct ratio  $> 0.5$  → hemothorax; triglycerides  $> 110$  mg/dL → chylothorax; amylase elevation → pancreatitis or esophageal rupture; pleural eosinophilia → air/blood, drug reaction, asbestos.

### BOARD TRAP

ADA  $> 40$  strongly suggests TB pleuritis, and pleural fluid AFB smear/culture are LOW-yield (sensitivity often  $< 30$ – $40\%$ ) — the trap is ruling out TB based on a negative pleural culture. Use ADA (and pleural biopsy with granulomas) rather than relying on fluid cultures to exclude TB.

## 14. Chylothorax / hemothorax clues

### WHAT YOU SEE

Labels 'TRIGLYCERIDES → CHYLOTHORAX' and 'Hct → HEMOTHORAX'

### WHAT IT ENCODES

Chylothorax: milky fluid with triglycerides  $> 110$  mg/dL (or chylomicrons on lipoprotein analysis), from thoracic duct disruption — trauma/surgery or lymphoma. Hemothorax: pleural-fluid/blood hematocrit ratio  $> 0.5$ , from trauma, malignancy, or PE with infarction; needs large-bore drainage and possibly surgery for high-volume bleeding.

### BOARD TRAP

Milky pleural fluid is not always chylous — pseudochylothorax (a chronic effusion, e.g., longstanding TB or rheumatoid) has HIGH cholesterol with cholesterol crystals but LOW triglycerides. The trap is calling every milky effusion a chylothorax; check triglycerides (high = chyle) vs cholesterol (high with low TG = pseudochylothorax).